Causal Analysis Report

bpic12-a.xes.gz-imf

Decision points: 41507

Choice sets: 6

Transitions fitted: 11

Significant (p<0.05): 2

Generated: 2026-06-24 09:05

Step 1 — Association test (logistic regression + permutation)
H□: choice □ util | choice_set.   reject_H0=True → util predicts routing.

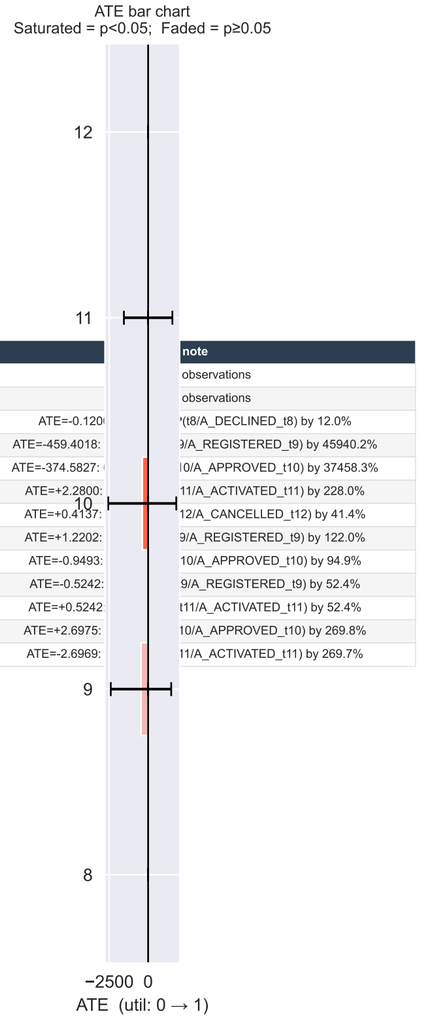
(no data)

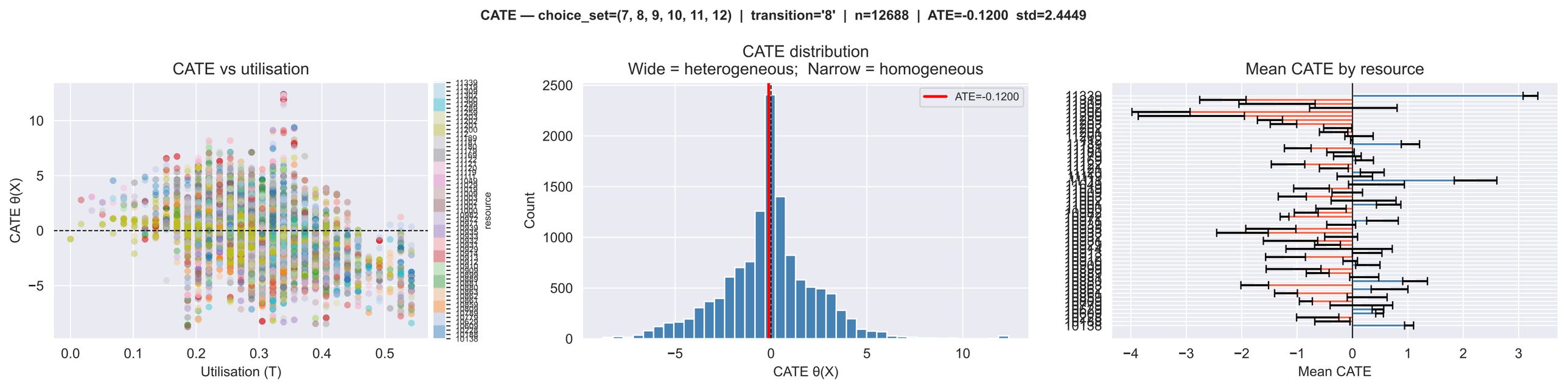
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

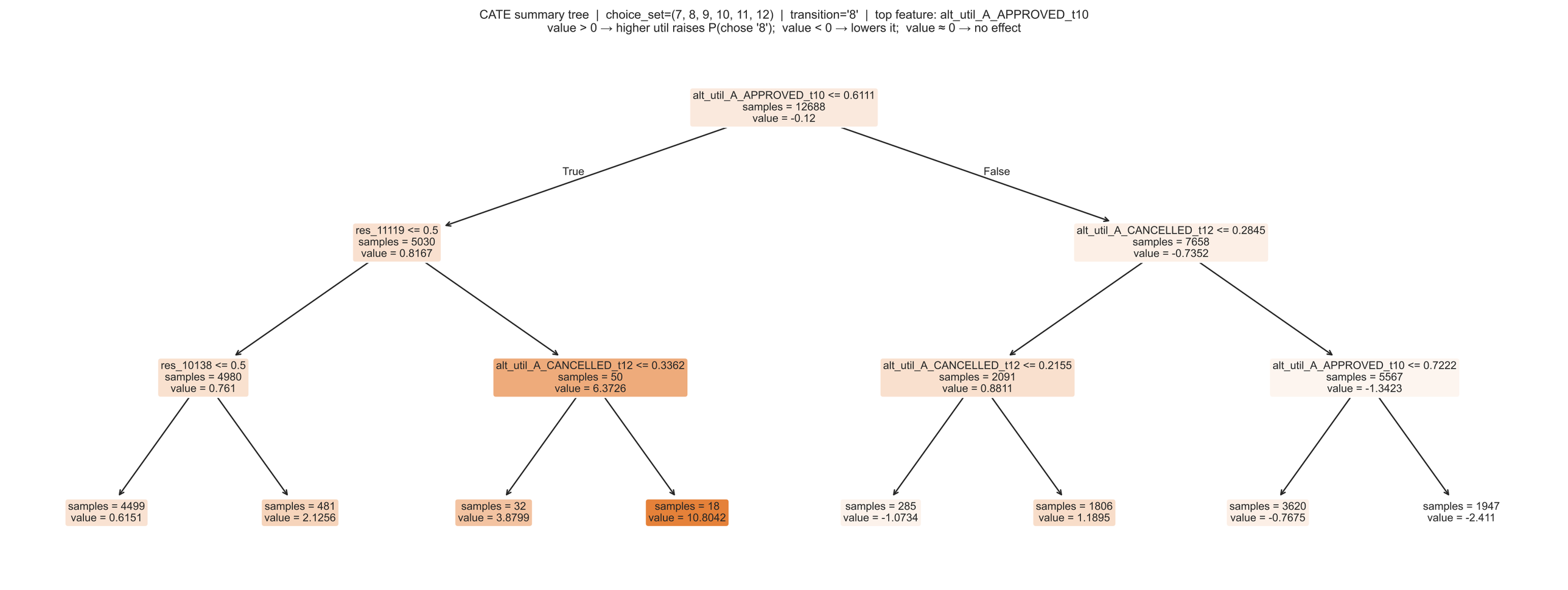
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(3, 4)	('τ_t3', 'A_ACCEPTED_t4')	5113	4	—	—	—	nan	nan	—	nan	—	too few observations
(5, 6)	('τ_t5', 'A_FINALIZED_t6')	5015	6	—	—	—	nan	nan	—	nan	—	too few observations
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	8	A_DECLINED_t8	A_ACTIVATED_t11, A_APPROVED_t10, A_CANCELLED_t12, A_REGISTERED_t9	0.215279	0.28318	[-0.3398, 0.7703]	0.4471	False	θ=+0.2153: 1-unit ↑ util ↑ P(t8/A_DECLINED_t8) by 21.5%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	9	A_REGISTERED_t9	A_ACTIVATED_t11, A_APPROVED_t10, A_CANCELLED_t12, A_DECLINED_t8	0.237885	2.682848	[-5.0205, 5.4963]	0.9293	False	θ=+0.2379: 1-unit ↑ util ↑ P(t9/A_REGISTERED_t9) by 23.8%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	10	A_APPROVED_t10	A_ACTIVATED_t11, A_CANCELLED_t12, A_DECLINED_t8, A_REGISTERED_t9	0.52222	3.021994	[-5.4009, 6.4453]	0.8628	False	θ=+0.5222: 1-unit ↑ util ↑ P(t10/A_APPROVED_t10) by 52.2%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	11	A_ACTIVATED_t11	A_APPROVED_t10, A_CANCELLED_t12, A_DECLINED_t8, A_REGISTERED_t9	-0.222601	2.346304	[-4.8214, 4.3762]	0.9244	False	θ=-0.2226: 1-unit ↑ util ↓ P(t11/A_ACTIVATED_t11) by 22.3%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	12	A_CANCELLED_t12	A_ACTIVATED_t11, A_APPROVED_t10, A_DECLINED_t8, A_REGISTERED_t9	0.494409	0.24941	[0.0056, 0.9833]	0.0474	True	θ=+0.4944: 1-unit ↑ util ↑ P(t12/A_CANCELLED_t12) by 49.4%
(9, 10)	('A_REGISTERED_t9', 'A_APPROVED_t10')	476	9, 10	9	A_REGISTERED_t9	A_APPROVED_t10	0.442707	0.904219	[-1.3296, 2.2150]	0.6244	False	θ=+0.4427: 1-unit ↑ util ↑ P(t9/A_REGISTERED_t9) by 44.3%
(9, 10)	('A_REGISTERED_t9', 'A_APPROVED_t10')	476	9, 10	10	A_APPROVED_t10	A_REGISTERED_t9	-0.442707	0.904219	[-2.2150, 1.3296]	0.6244	False	θ=-0.4427: 1-unit ↑ util ↓ P(t10/A_APPROVED_t10) by 44.3%
(9, 11)	('A_REGISTERED_t9', 'A_ACTIVATED_t11')	1055	9, 11	9	A_REGISTERED_t9	A_ACTIVATED_t11	-0.302604	1.236002	[-2.7252, 2.1200]	0.8066	False	θ=-0.3026: 1-unit ↑ util ↓ P(t9/A_REGISTERED_t9) by 30.3%
(9, 11)	('A_REGISTERED_t9', 'A_ACTIVATED_t11')	1055	9, 11	11	A_ACTIVATED_t11	A_REGISTERED_t9	0.302604	1.236002	[-2.1200, 2.7252]	0.8066	False	θ=+0.3026: 1-unit ↑ util ↑ P(t11/A_ACTIVATED_t11) by 30.3%
(10, 11)	('A_APPROVED_t10', 'A_ACTIVATED_t11')	715	10, 11	10	A_APPROVED_t10	A_ACTIVATED_t11	0.155171	0.957591	[-1.7217, 2.0320]	0.8713	False	θ=+0.1552: 1-unit ↑ util ↑ P(t10/A_APPROVED_t10) by 15.5%
(10, 11)	('A_APPROVED_t10', 'A_ACTIVATED_t11')	715	10, 11	11	A_ACTIVATED_t11	A_APPROVED_t10	-0.155171	0.957591	[-2.0320, 1.7217]	0.8713	False	θ=-0.1552: 1-unit ↑ util ↓ P(t11/A_ACTIVATED_t11) by 15.5%

Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

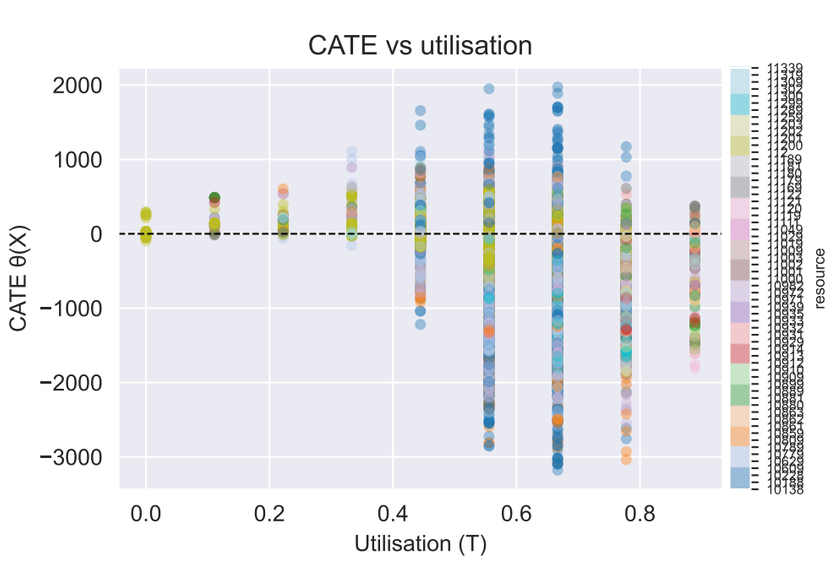
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0		note
(3, 4)	('τ_t3', 'A_ACCEPTED_t4')	5113	4	—	—	—	nan	—	nan	—		observations
(5, 6)	('τ_t5', 'A_FINALIZED_t6')	5015	6	—	—	—	nan	—	nan	—		observations
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	8	A_DECLINED_t8	A_ACTIVATED_t11, A_APPROVED_t10, A_CANCELLED_t12, A_REGISTERED_t9	-0.119959	[-4.1715, 3.9316]	0.9537	False	ATE=-0.120	'(t8/A_DECLINED_t8) by 12.0%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	9	A_REGISTERED_t9	A_ACTIVATED_t11, A_APPROVED_t10, A_CANCELLED_t12, A_DECLINED_t8	-459.401827	[-2404.5658, 1485.7621]	0.6434	False	ATE=-459.4018:	9/A_REGISTERED_t9) by 45940.2%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	10	A_APPROVED_t10	A_ACTIVATED_t11, A_CANCELLED_t12, A_DECLINED_t8, A_REGISTERED_t9	-374.582741	[-2563.2998, 1814.1343]	0.7373	False	ATE=-374.5827:	10/A_APPROVED_t10) by 37458.3%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	11	A_ACTIVATED_t11	A_APPROVED_t10, A_CANCELLED_t12, A_DECLINED_t8, A_REGISTERED_t9	2.27997	[-1558.8803, 1563.4402]	0.9977	False	ATE=+2.2800:	11/A_ACTIVATED_t11) by 228.0%
(7, 8, 9, 10, 11, 12)	('τ_t7', 'A_DECLINED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_ACTIVATED_t11', 'A_CANCELLED_t12')	12688	8, 9, 10, 11, 12	12	A_CANCELLED_t12	A_ACTIVATED_t11, A_APPROVED_t10, A_DECLINED_t8, A_REGISTERED_t9	0.413697	[-3.1226, 3.9500]	0.8186	False	ATE=+0.4137:	12/A_CANCELLED_t12) by 41.4%
(9, 10)	('A_REGISTERED_t9', 'A_APPROVED_t10')	476	9, 10	9	A_REGISTERED_t9	A_APPROVED_t10	1.220177	[-3.7505, 6.1909]	0.6304	False	ATE=+1.2202:	9/A_REGISTERED_t9) by 122.0%
(9, 10)	('A_REGISTERED_t9', 'A_APPROVED_t10')	476	9, 10	10	A_APPROVED_t10	A_REGISTERED_t9	-0.949345	[-5.4434, 3.5447]	0.6788	False	ATE=-0.9493:	10/A_APPROVED_t10) by 94.9%
(9, 11)	('A_REGISTERED_t9', 'A_ACTIVATED_t11')	1055	9, 11	9	A_REGISTERED_t9	A_ACTIVATED_t11	-0.524243	[-2.3980, 1.3495]	0.5834	False	ATE=-0.5242:	9/A_REGISTERED_t9) by 52.4%
(9, 11)	('A_REGISTERED_t9', 'A_ACTIVATED_t11')	1055	9, 11	11	A_ACTIVATED_t11	A_REGISTERED_t9	0.524168	[-1.3493, 2.3976]	0.5834	False	ATE=+0.5242	t11/A_ACTIVATED_t11) by 52.4%
(10, 11)	('A_APPROVED_t10', 'A_ACTIVATED_t11')	715	10, 11	10	A_APPROVED_t10	A_ACTIVATED_t11	2.697526	[0.1805, 5.2146]	0.0357	True	ATE=+2.6975:	10/A_APPROVED_t10) by 269.8%
(10, 11)	('A_APPROVED_t10', 'A_ACTIVATED_t11')	715	10, 11	11	A_ACTIVATED_t11	A_APPROVED_t10	-2.696851	[-5.2185, -0.1752]	0.0361	True	ATE=-2.6969:	11/A_ACTIVATED_t11) by 269.7%

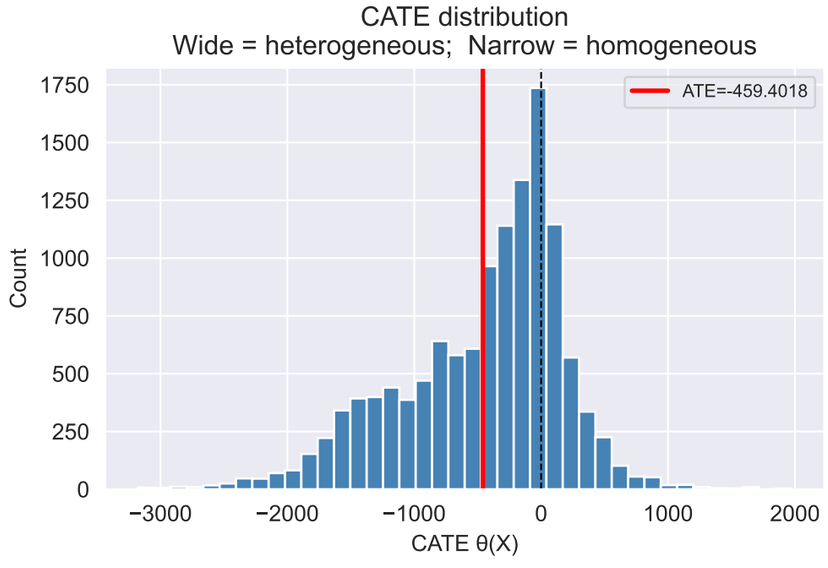


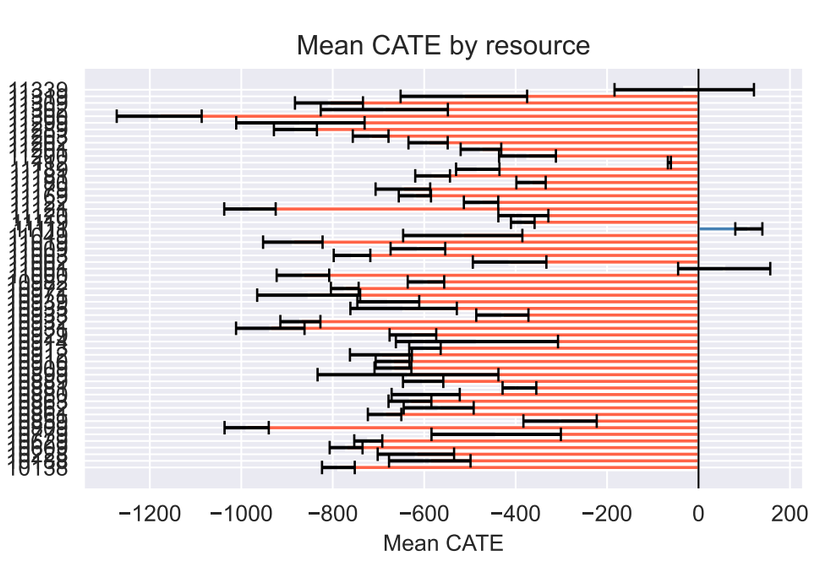

CATE — choice_set=(7, 8, 9, 10, 11, 12) | transition='8' | n=12688 | ATE=-0.1200 std=2.4449
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1200
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

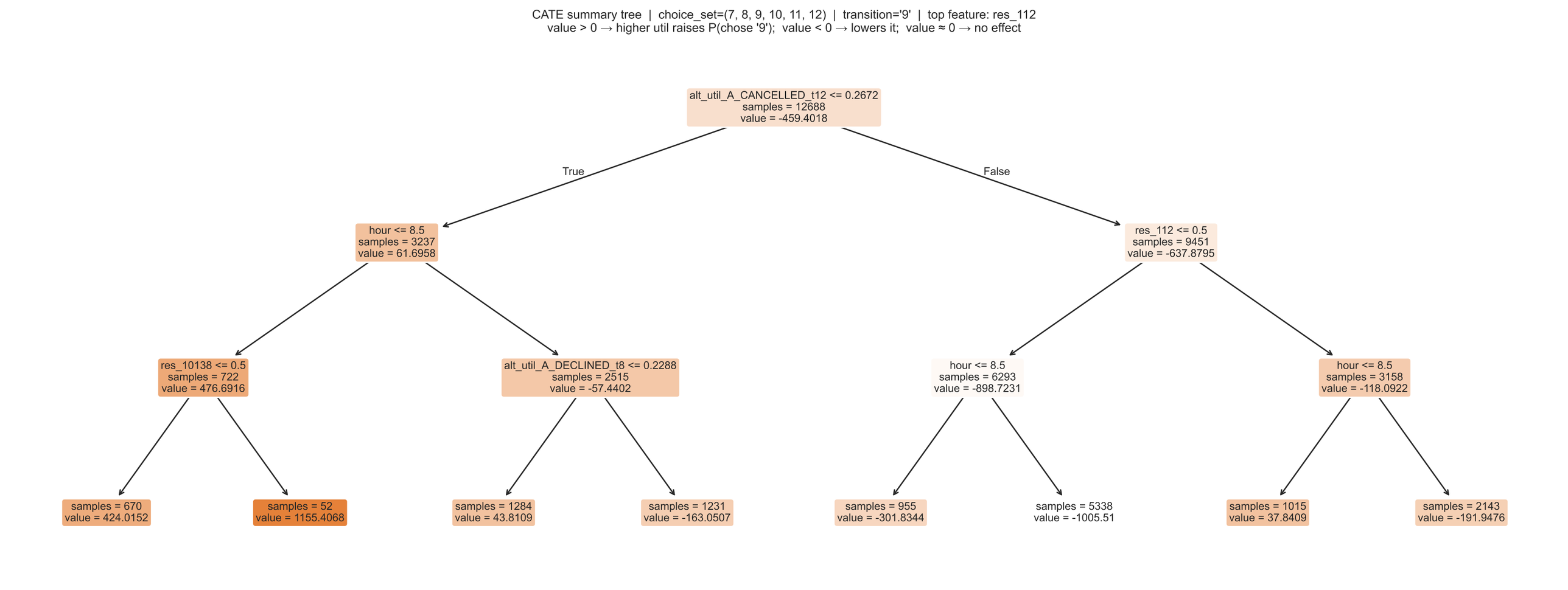

CATE summary tree | choice_set=(7, 8, 9, 10, 11, 12) | transition='8' | top feature: alt_util_A_APPROVED_t10
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_APPROVED_t10 <= 0.6111
samples = 12688
value = -0.12
True
False
res_11119 <= 0.5
samples = 5030
value = 0.8167
alt_util_A_CANCELLED_t12 <= 0.2845
samples = 7658
value = -0.7352
res_10138 <= 0.5
samples = 4980
value = 0.761
alt_util_A_CANCELLED_t12 <= 0.3362
samples = 50
value = 6.3726
alt_util_A_CANCELLED_t12 <= 0.2155
samples = 2091
value = 0.8811
alt_util_A_APPROVED_t10 <= 0.7222
samples = 5567
value = -1.3423
samples = 4499
value = 0.6151
samples = 481
value = 2.1256
samples = 32
value = 3.8799
samples = 18
value = 10.8042
samples = 285
value = -1.0734
samples = 1806
value = 1.1895
samples = 3620
value = -0.7675
samples = 1947
value = -2.411

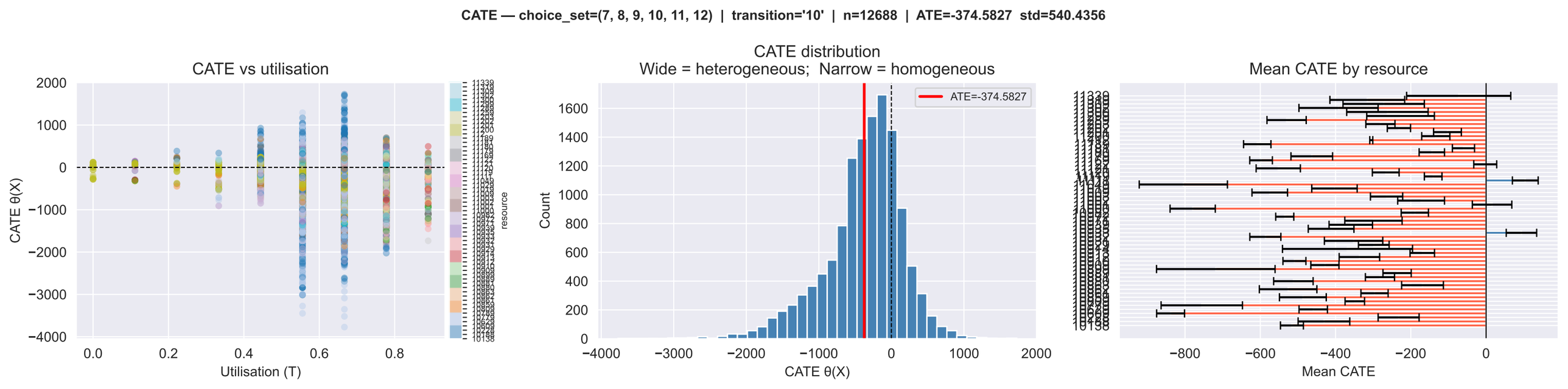
CATE — choice_set=(7, 8, 9, 10, 11, 12) | transition='9' | n=12688 | ATE=-459.4018 std=651.9562

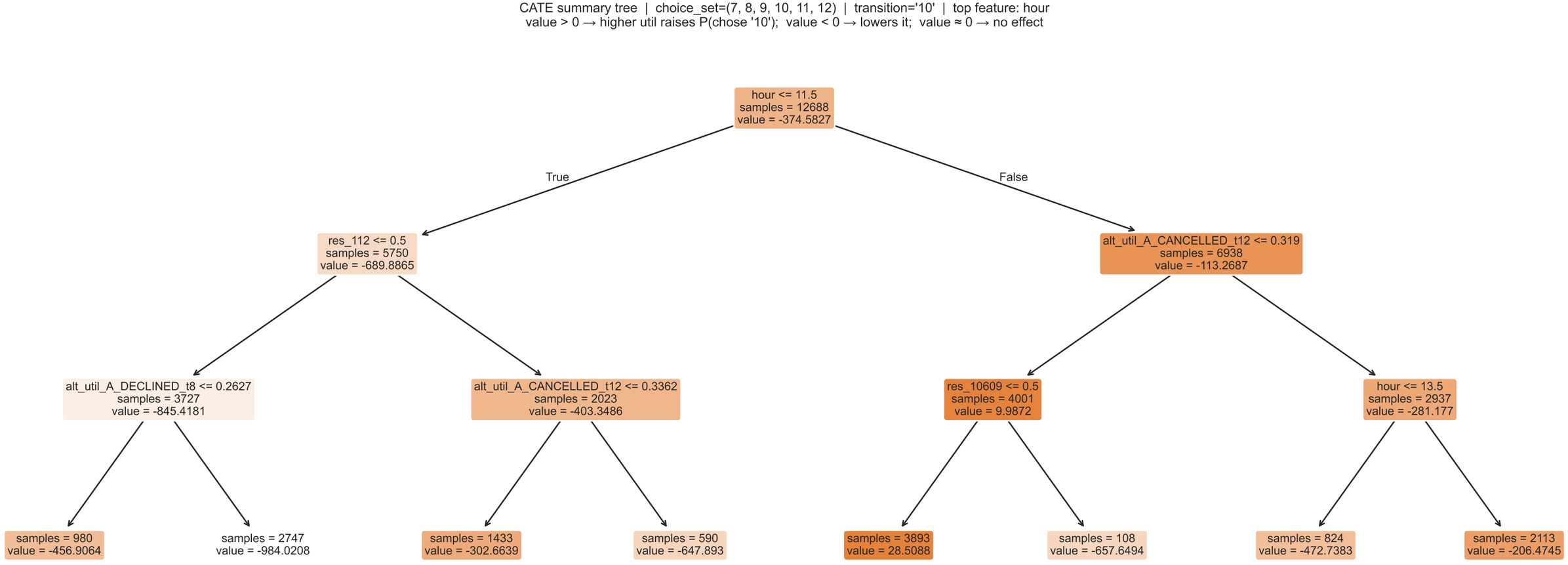




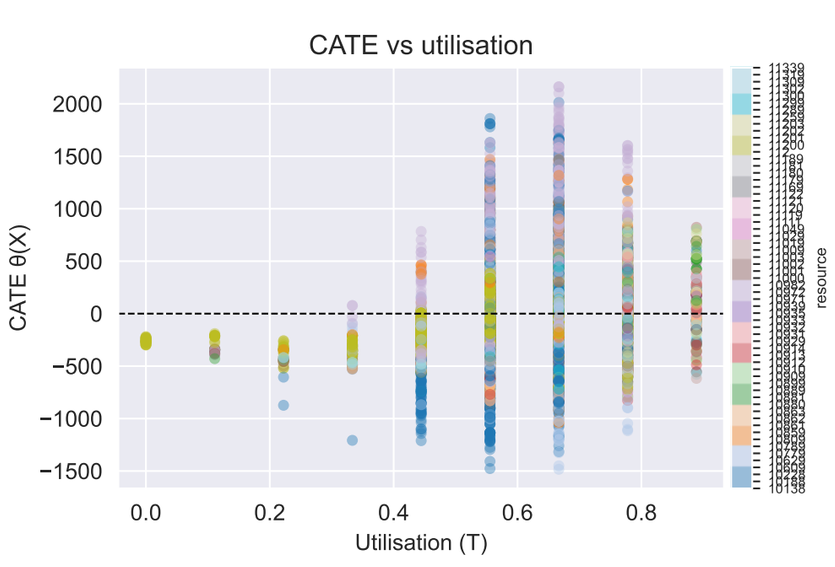


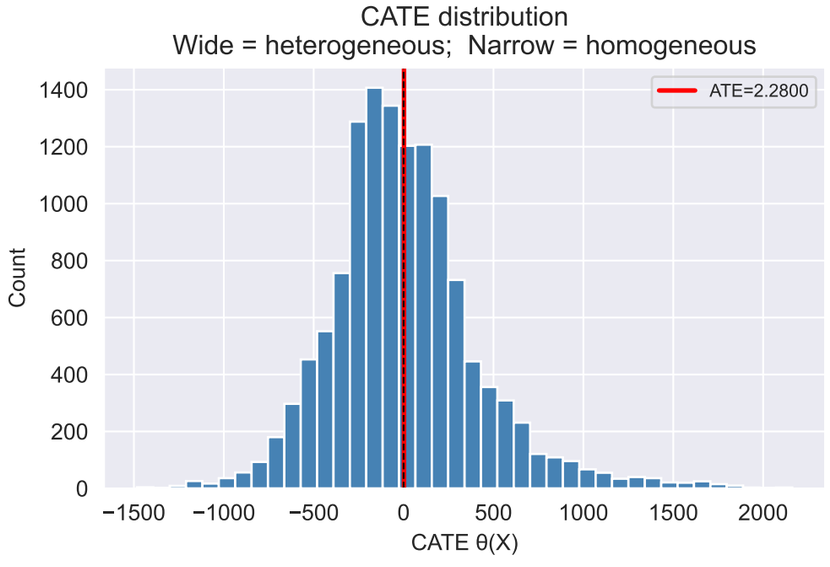

CATE summary tree | choice_set=(7, 8, 9, 10, 11, 12) | transition='9' | top feature: res_112
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_CANCELLED_t12 <= 0.2672
samples = 12688
value = -459.4018
True
False
hour <= 8.5
samples = 3237
value = 61.6958
res_112 <= 0.5
samples = 9451
value = -637.8795
res_10138 <= 0.5
samples = 722
value = 476.6916
alt_util_A_DECLINED_t8 <= 0.2288
samples = 2515
value = -57.4402
hour <= 8.5
samples = 6293
value = -898.7231
hour <= 8.5
samples = 3158
value = -118.0922
samples = 670
value = 424.0152
samples = 52
value = 1155.4068
samples = 1284
value = 43.8109
samples = 1231
value = -163.0507
samples = 955
value = -301.8344
samples = 5338
value = -1005.51
samples = 1015
value = 37.8409
samples = 2143
value = -191.9476

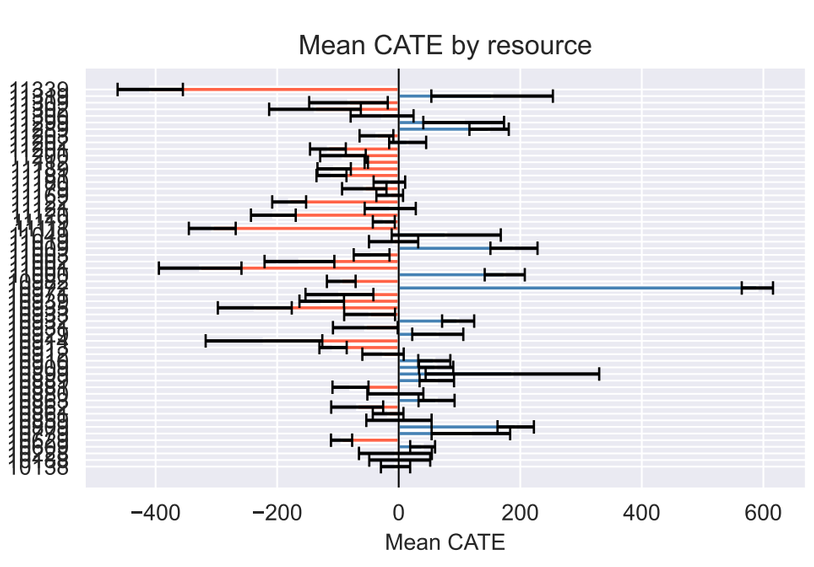

CATE — choice_set=(7, 8, 9, 10, 11, 12) | transition='10' | n=12688 | ATE=-374.5827 std=540.4356
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-374.5827
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

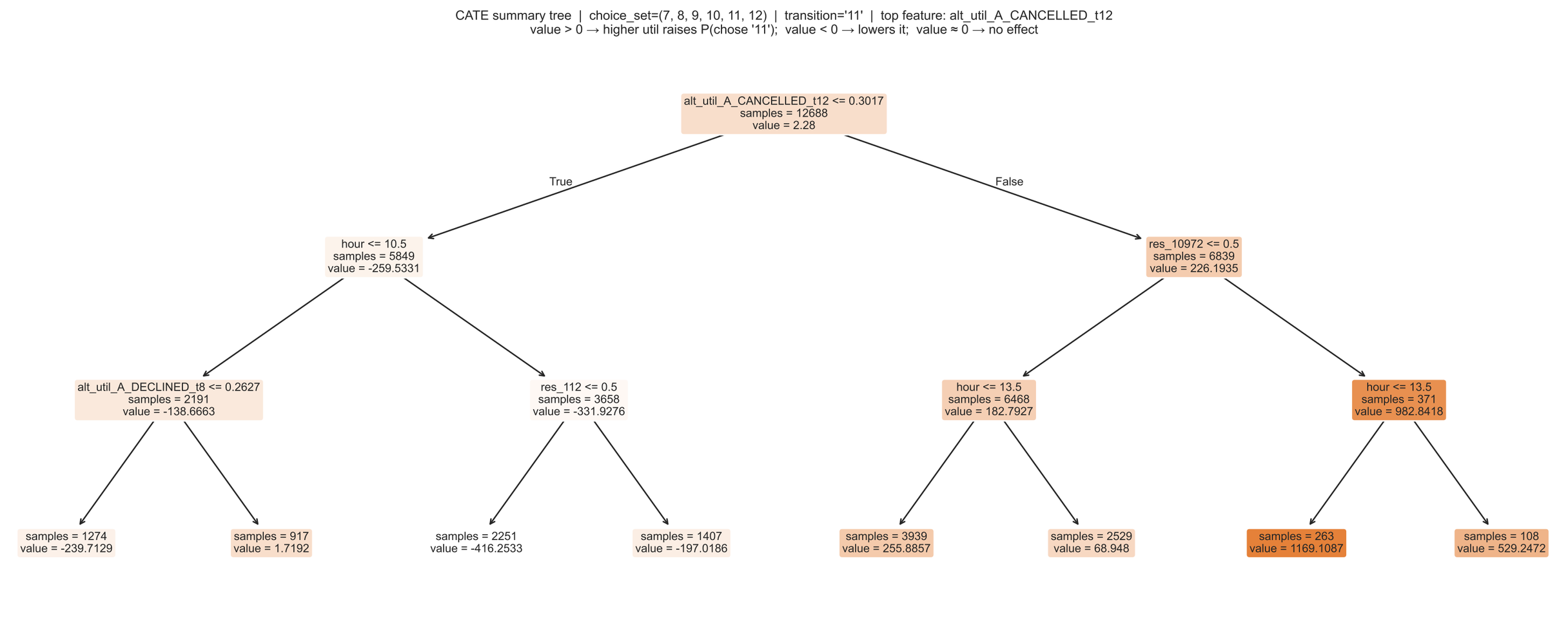

CATE summary tree | choice_set=(7, 8, 9, 10, 11, 12) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 11.5
samples = 12688
value = -374.5827
True
False
res_112 <= 0.5
samples = 5750
value = -689.8865
alt_util_A_CANCELLED_t12 <= 0.319
samples = 6938
value = -113.2687
alt_util_A_DECLINED_t8 <= 0.2627
samples = 3727
value = -845.4181
alt_util_A_CANCELLED_t12 <= 0.3362
samples = 2023
value = -403.3486
res_10609 <= 0.5
samples = 4001
value = 9.9872
hour <= 13.5
samples = 2937
value = -281.177
samples = 980
value = -456.9064
samples = 2747
value = -984.0208
samples = 1433
value = -302.6639
samples = 590
value = -647.893
samples = 3893
value = 28.5088
samples = 108
value = -657.6494
samples = 824
value = -472.7383
samples = 2113
value = -206.4745

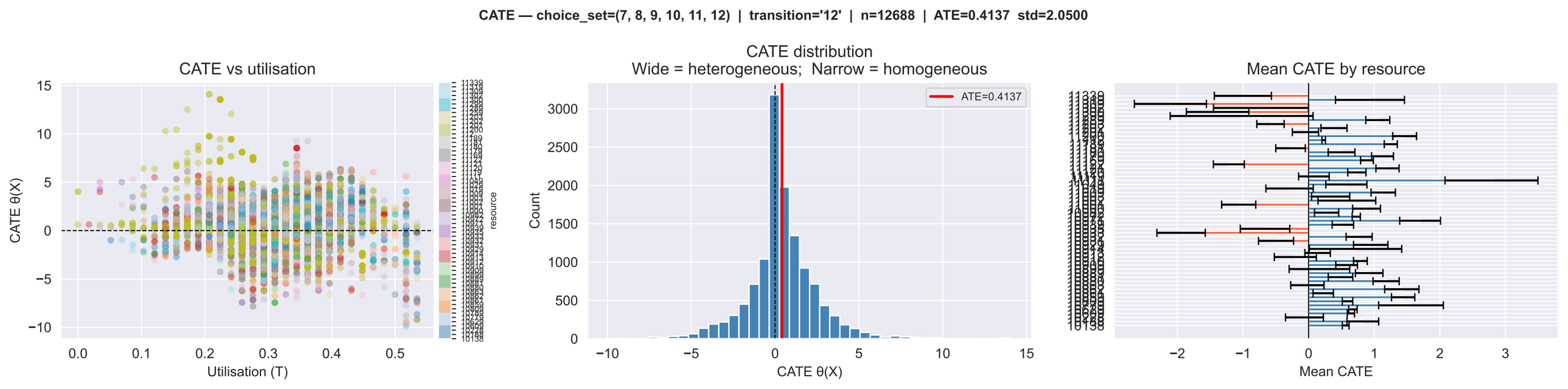
CATE — choice_set=(7, 8, 9, 10, 11, 12) | transition='11' | n=12688 | ATE=2.2800 std=412.3961

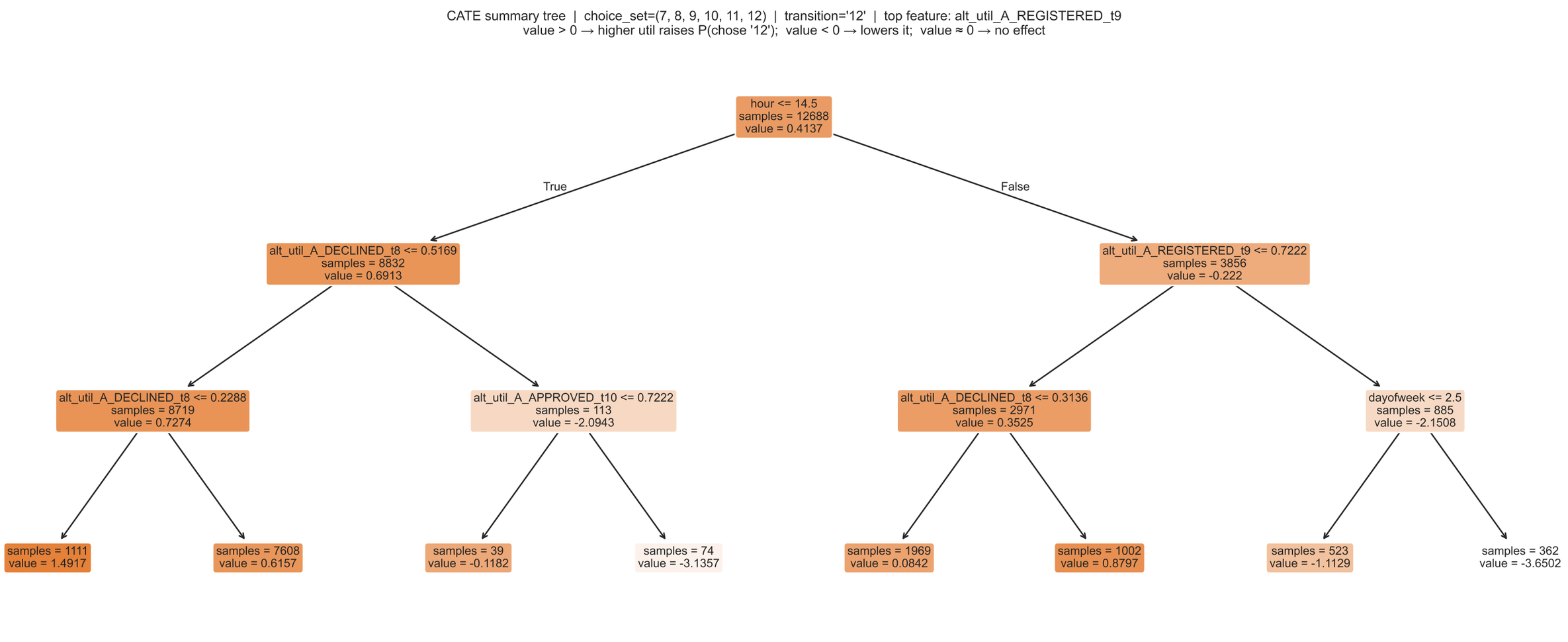


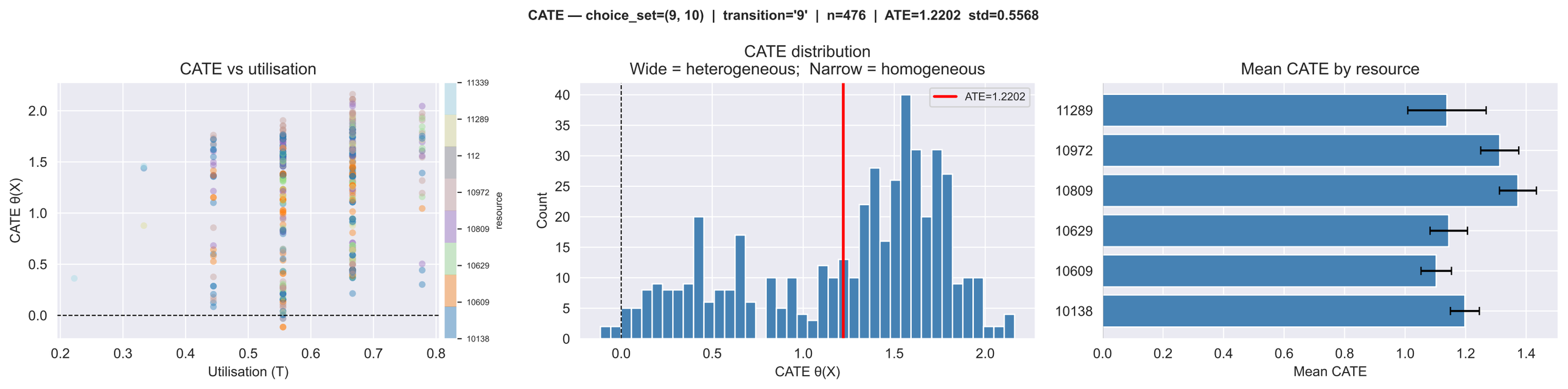


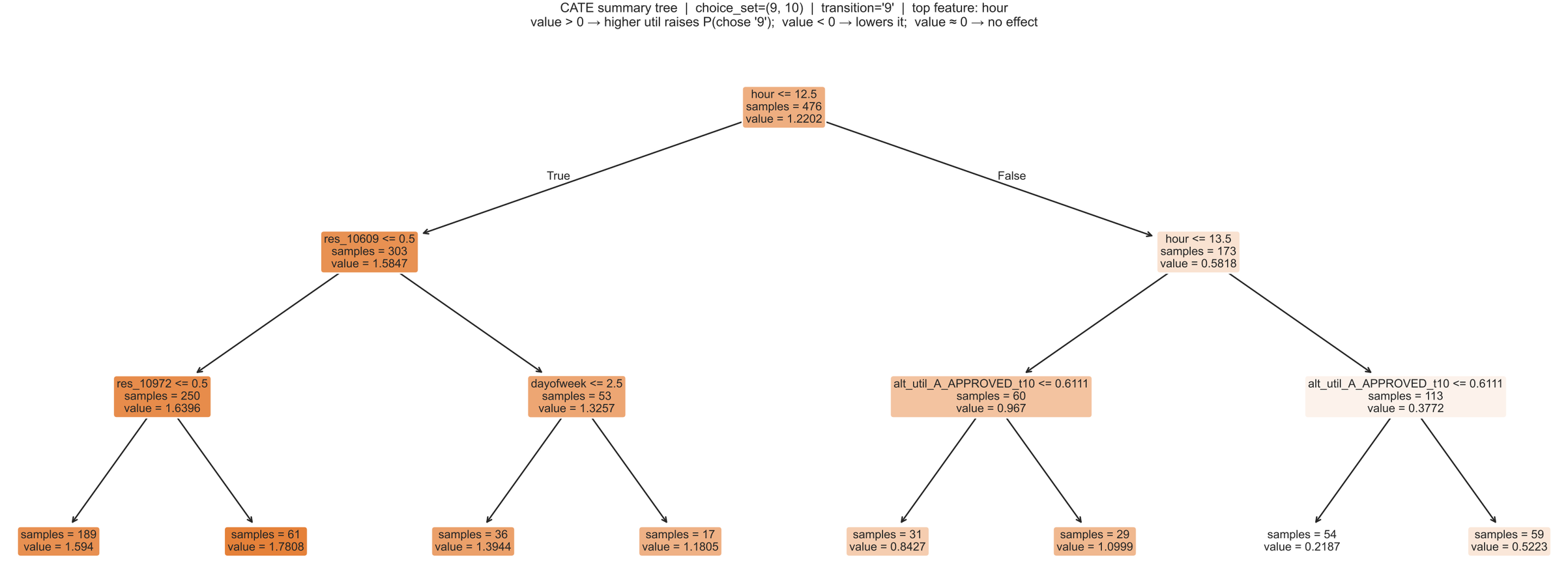


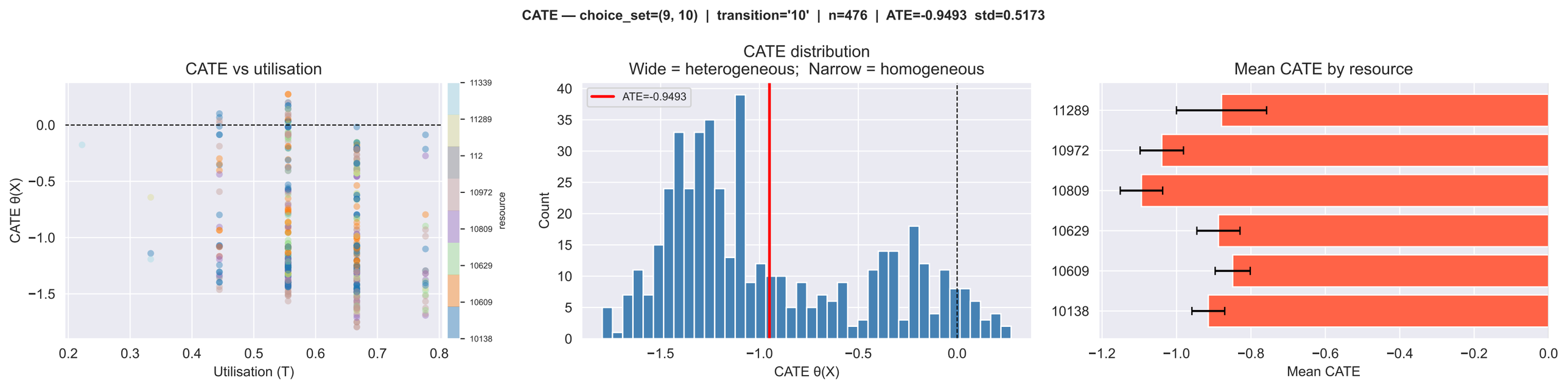

CATE summary tree | choice_set=(7, 8, 9, 10, 11, 12) | transition='11' | top feature: alt_util_A_CANCELLED_t12
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_CANCELLED_t12 <= 0.3017
samples = 12688
value = 2.28
True
False
hour <= 10.5
samples = 5849
value = -259.5331
res_10972 <= 0.5
samples = 6839
value = 226.1935
alt_util_A_DECLINED_t8 <= 0.2627
samples = 2191
value = -138.6663
res_112 <= 0.5
samples = 3658
value = -331.9276
hour <= 13.5
samples = 6468
value = 182.7927
hour <= 13.5
samples = 371
value = 982.8418
samples = 1274
value = -239.7129
samples = 917
value = 1.7192
samples = 2251
value = -416.2533
samples = 1407
value = -197.0186
samples = 3939
value = 255.8857
samples = 2529
value = 68.948
samples = 263
value = 1169.1087
samples = 108
value = 529.2472

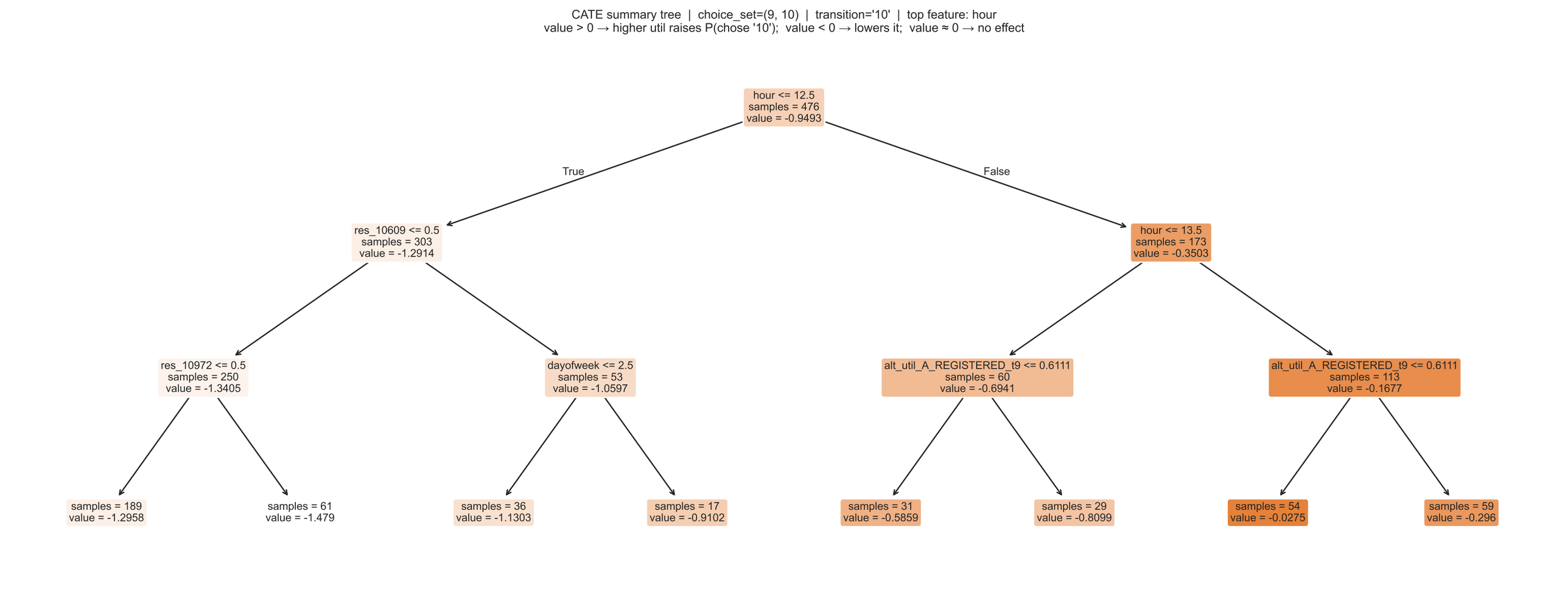

CATE — choice_set=(7, 8, 9, 10, 11, 12) | transition='12' | n=12688 | ATE=0.4137 std=2.0500
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.4137
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

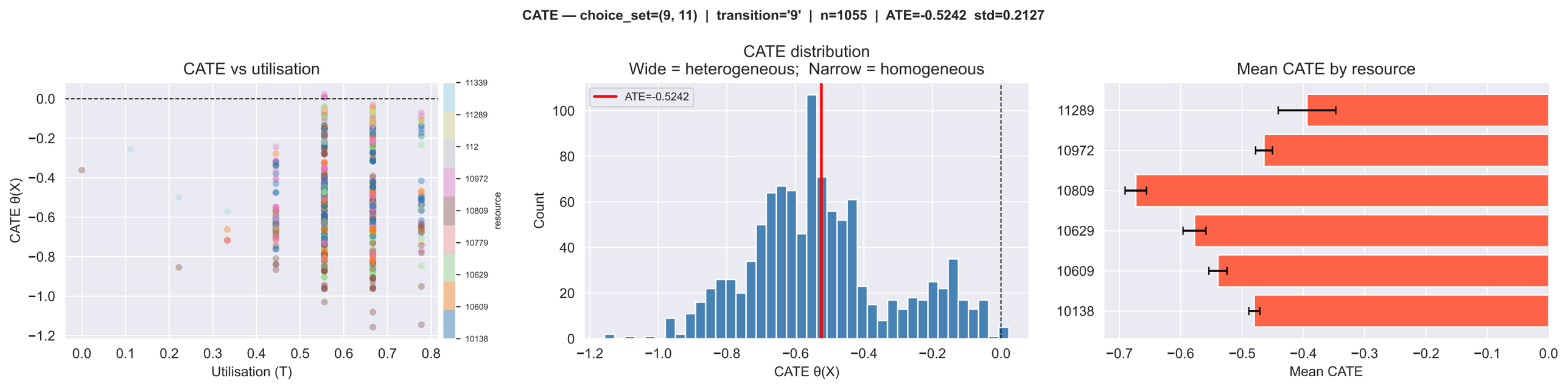

CATE summary tree | choice_set=(7, 8, 9, 10, 11, 12) | transition='12' | top feature: alt_util_A_REGISTERED_t9
value > 0 → higher util raises P(chose '12'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 14.5
samples = 12688
value = 0.4137
True
False
alt_util_A_DECLINED_t8 <= 0.5169
samples = 8832
value = 0.6913
alt_util_A_REGISTERED_t9 <= 0.7222
samples = 3856
value = -0.222
alt_util_A_DECLINED_t8 <= 0.2288
samples = 8719
value = 0.7274
alt_util_A_APPROVED_t10 <= 0.7222
samples = 113
value = -2.0943
alt_util_A_DECLINED_t8 <= 0.3136
samples = 2971
value = 0.3525
dayofweek <= 2.5
samples = 885
value = -2.1508
samples = 1111
value = 1.4917
samples = 7608
value = 0.6157
samples = 39
value = -0.1182
samples = 74
value = -3.1357
samples = 1969
value = 0.0842
samples = 1002
value = 0.8797
samples = 523
value = -1.1129
samples = 362
value = -3.6502

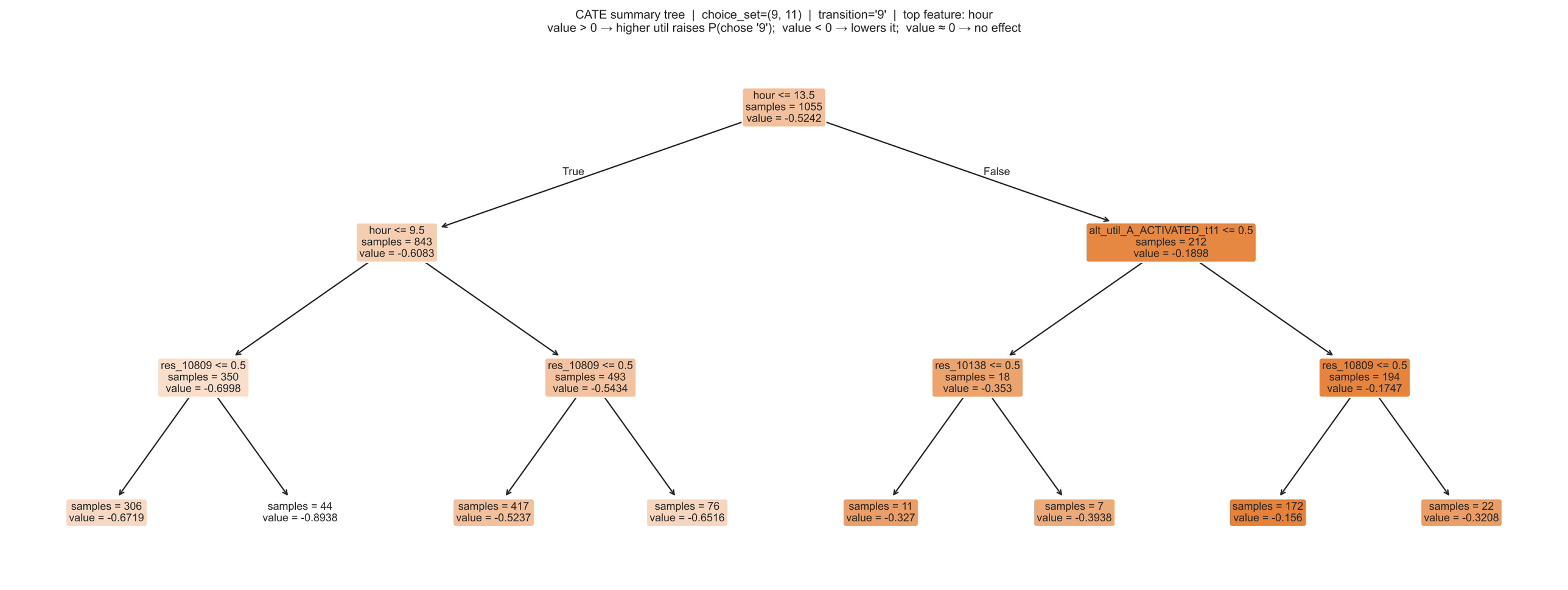

CATE — choice_set=(9, 10) | transition='9' | n=476 | ATE=1.2202 std=0.5568
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=1.2202
Count
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE

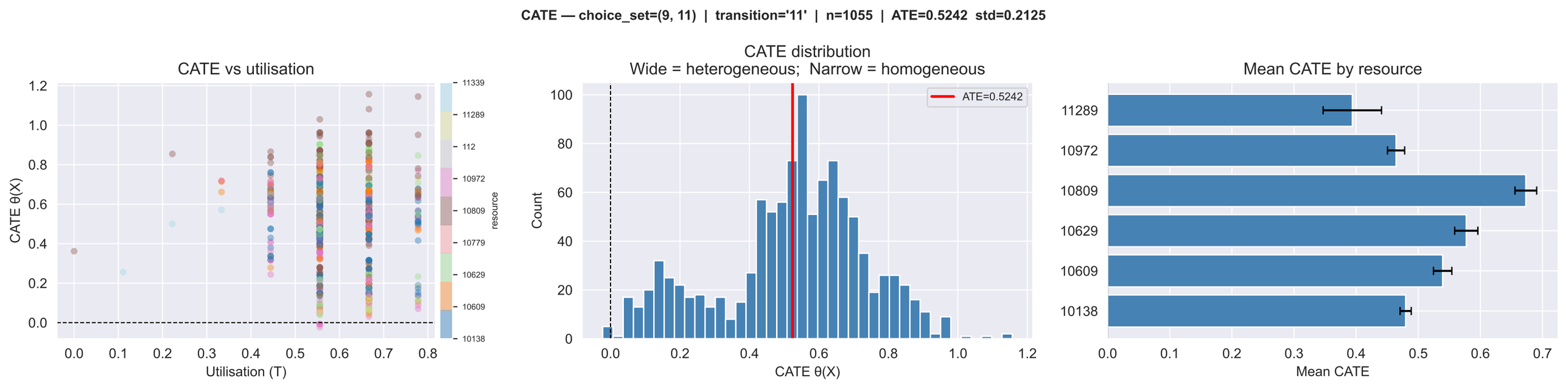

CATE summary tree | choice_set=(9, 10) | transition='9' | top feature: hour
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 12.5
samples = 476
value = 1.2202
True
False
res_10609 <= 0.5
samples = 303
value = 1.5847
hour <= 13.5
samples = 173
value = 0.5818
res_10972 <= 0.5
samples = 250
value = 1.6396
dayofweek <= 2.5
samples = 53
value = 1.3257
alt_util_A_APPROVED_t10 <= 0.6111
samples = 60
value = 0.967
alt_util_A_APPROVED_t10 <= 0.6111
samples = 113
value = 0.3772
samples = 189
value = 1.594
samples = 61
value = 1.7808
samples = 36
value = 1.3944
samples = 17
value = 1.1805
samples = 31
value = 0.8427
samples = 29
value = 1.0999
samples = 54
value = 0.2187
samples = 59
value = 0.5223

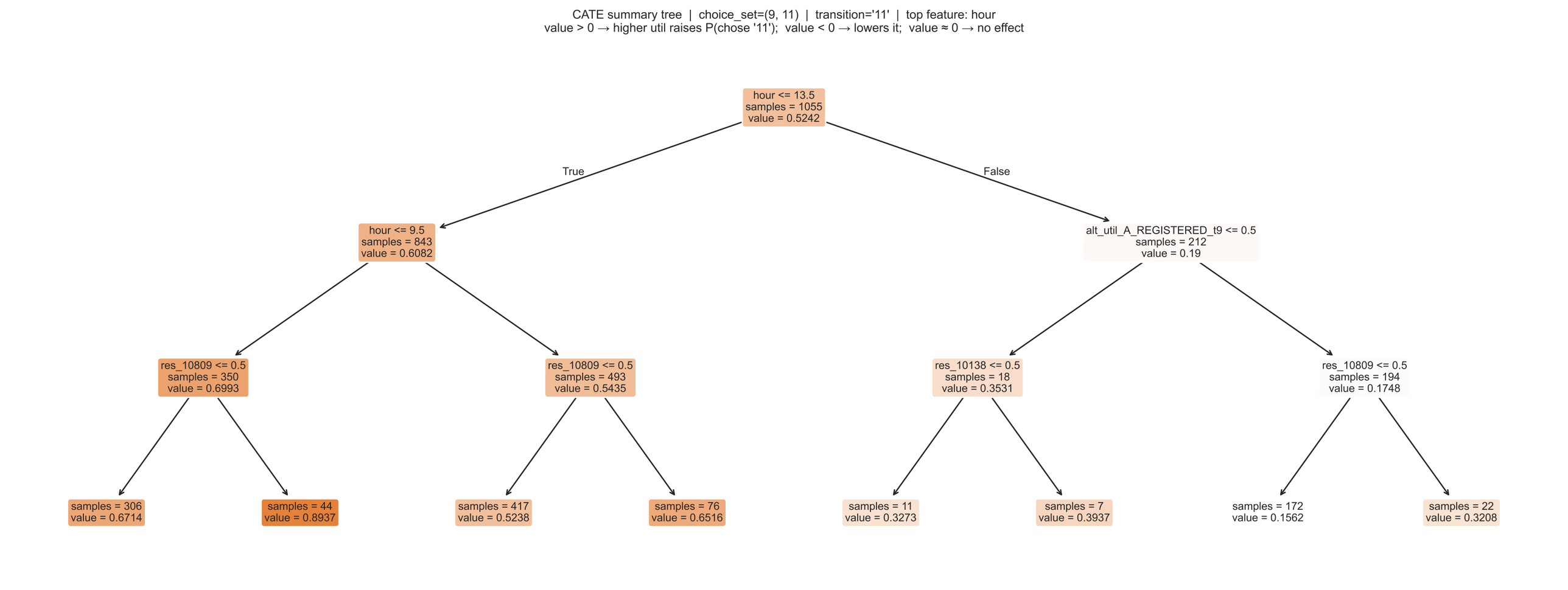

CATE — choice_set=(9, 10) | transition='10' | n=476 | ATE=-0.9493 std=0.5173
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.9493
Count
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE

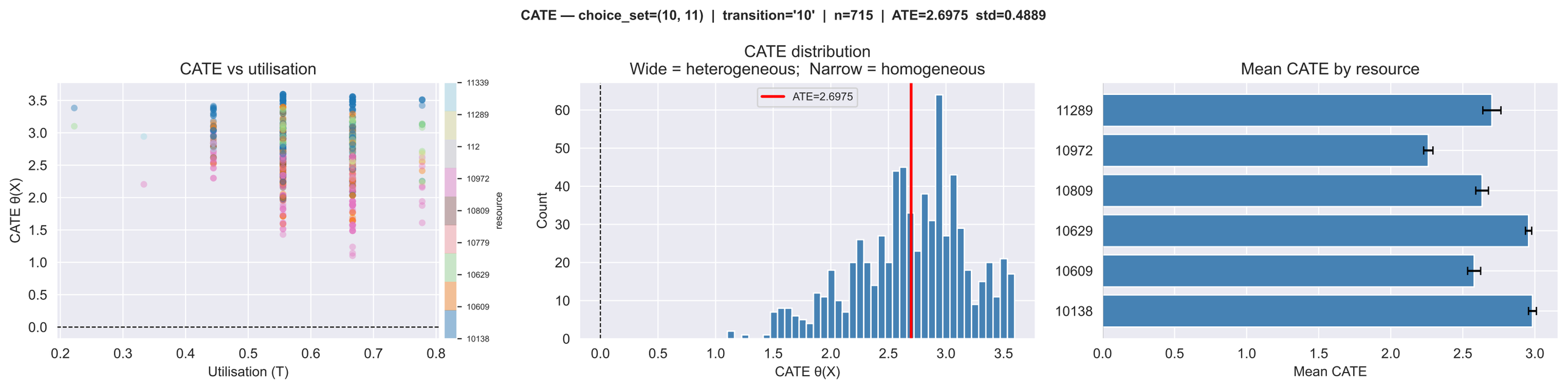

CATE summary tree | choice_set=(9, 10) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 12.5
samples = 476
value = -0.9493
True
False
res_10609 <= 0.5
samples = 303
value = -1.2914
hour <= 13.5
samples = 173
value = -0.3503
res_10972 <= 0.5
samples = 250
value = -1.3405
dayofweek <= 2.5
samples = 53
value = -1.0597
alt_util_A_REGISTERED_t9 <= 0.6111
samples = 60
value = -0.6941
alt_util_A_REGISTERED_t9 <= 0.6111
samples = 113
value = -0.1677
samples = 189
value = -1.2958
samples = 61
value = -1.479
samples = 36
value = -1.1303
samples = 17
value = -0.9102
samples = 31
value = -0.5859
samples = 29
value = -0.8099
samples = 54
value = -0.0275
samples = 59
value = -0.296

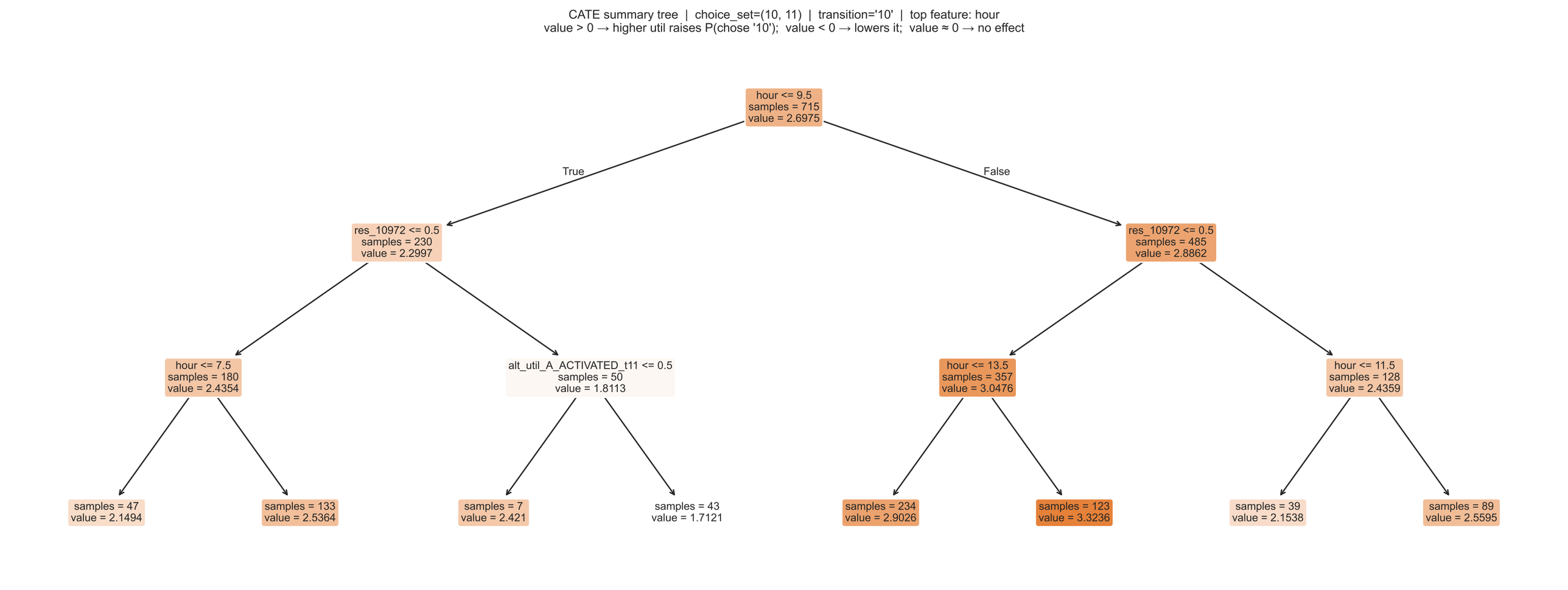

CATE — choice_set=(9, 11) | transition='9' | n=1055 | ATE=-0.5242 std=0.2127
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10779
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.5242
Count
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE

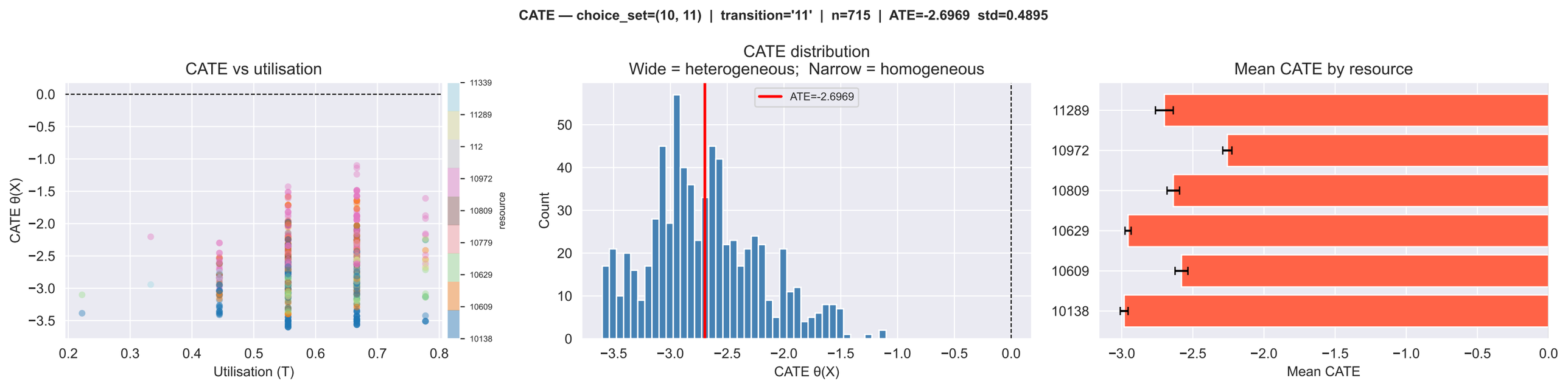

CATE summary tree | choice_set=(9, 11) | transition='9' | top feature: hour
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 13.5
samples = 1055
value = -0.5242
True
False
hour <= 9.5
samples = 843
value = -0.6083
alt_util_A_ACTIVATED_t11 <= 0.5
samples = 212
value = -0.1898
res_10809 <= 0.5
samples = 350
value = -0.6998
res_10809 <= 0.5
samples = 493
value = -0.5434
res_10138 <= 0.5
samples = 18
value = -0.353
res_10809 <= 0.5
samples = 194
value = -0.1747
samples = 306
value = -0.6719
samples = 44
value = -0.8938
samples = 417
value = -0.5237
samples = 76
value = -0.6516
samples = 11
value = -0.327
samples = 7
value = -0.3938
samples = 172
value = -0.156
samples = 22
value = -0.3208

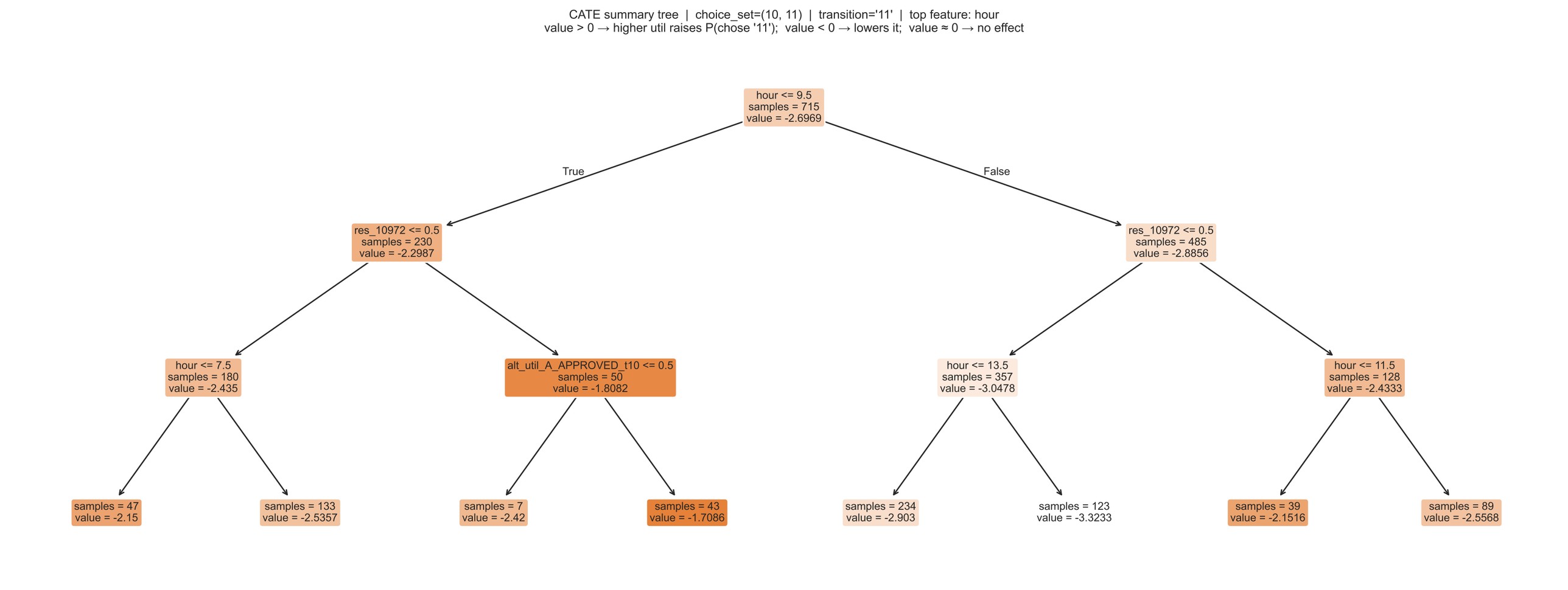

CATE — choice_set=(9, 11) | transition='11' | n=1055 | ATE=0.5242 std=0.2125
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10779
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
ATE=0.5242
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE


CATE summary tree | choice_set=(9, 11) | transition='11' | top feature: hour
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 13.5
samples = 1055
value = 0.5242
True
False
hour <= 9.5
samples = 843
value = 0.6082
alt_util_A_REGISTERED_t9 <= 0.5
samples = 212
value = 0.19
res_10809 <= 0.5
samples = 350
value = 0.6993
res_10809 <= 0.5
samples = 493
value = 0.5435
res_10138 <= 0.5
samples = 18
value = 0.3531
res_10809 <= 0.5
samples = 194
value = 0.1748
samples = 306
value = 0.6714
samples = 44
value = 0.8937
samples = 417
value = 0.5238
samples = 76
value = 0.6516
samples = 11
value = 0.3273
samples = 7
value = 0.3937
samples = 172
value = 0.1562
samples = 22
value = 0.3208


CATE — choice_set=(10, 11) | transition='10' | n=715 | ATE=2.6975 std=0.4889
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10779
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=2.6975
Count
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE


CATE summary tree | choice_set=(10, 11) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 9.5
samples = 715
value = 2.6975
True
False
res_10972 <= 0.5
samples = 230
value = 2.2997
res_10972 <= 0.5
samples = 485
value = 2.8862
hour <= 7.5
samples = 180
value = 2.4354
alt_util_A_ACTIVATED_t11 <= 0.5
samples = 50
value = 1.8113
hour <= 13.5
samples = 357
value = 3.0476
hour <= 11.5
samples = 128
value = 2.4359
samples = 47
value = 2.1494
samples = 133
value = 2.5364
samples = 7
value = 2.421
samples = 43
value = 1.7121
samples = 234
value = 2.9026
samples = 123
value = 3.3236
samples = 39
value = 2.1538
samples = 89
value = 2.5595


CATE — choice_set=(10, 11) | transition='11' | n=715 | ATE=-2.6969 std=0.4895
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10779
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.6969
Count
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE


CATE summary tree | choice_set=(10, 11) | transition='11' | top feature: hour
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 9.5
samples = 715
value = -2.6969
True
False
res_10972 <= 0.5
samples = 230
value = -2.2987
res_10972 <= 0.5
samples = 485
value = -2.8856
hour <= 7.5
samples = 180
value = -2.435
alt_util_A_APPROVED_t10 <= 0.5
samples = 50
value = -1.8082
hour <= 13.5
samples = 357
value = -3.0478
hour <= 11.5
samples = 128
value = -2.4333
samples = 47
value = -2.15
samples = 133
value = -2.5357
samples = 7
value = -2.42
samples = 43
value = -1.7086
samples = 234
value = -2.903
samples = 123
value = -3.3233
samples = 39
value = -2.1516
samples = 89
value = -2.5568

Big Data Analytics

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Amazon Nitro System · Google Cloud Titanium · Azure Boost · Amazon Elastic Fabric Adapter · Google Cloud Jupiter · Azure InfiniBand · Amazon Placement Groups · Google Cloud Compact Placement Policy · Azure Proximity Placement Groups · Amazon Local Zones · Google Cloud Local SSD · Azure Ultra Disk · Amazon Elastic Block Store · Google Cloud Persistent Disk · Azure Managed Disks · Amazon Instance Store · Google Cloud Local SSD · Azure Temporary Disk · Amazon Elastic File System · Google Cloud Filestore · Azure Files · Amazon FSx for Lustre · Google Cloud Parallelstore · Azure HPC Cache · Amazon FSx for Windows File Server · Google Cloud NetApp Volumes · Azure NetApp Files · Amazon FSx for NetApp ONTAP · Google Cloud NetApp Volumes · Azure NetApp Files · Amazon FSx for OpenZFS · Google Cloud Filestore · Azure Files · Amazon S3 · Google Cloud Storage · Azure Blob Storage · Amazon Glacier · Google Cloud Archive Storage · Azure Archive Storage · Amazon Snowball · Google Cloud Transfer Appliance · Azure Data Box · Amazon Snowmobile · Google Cloud Transfer Service · Azure Import/Export · Amazon DataSync · Google Cloud Storage Transfer Service · Azure Data Factory · Amazon Database Migration Service · Google Cloud Database Migration Service · Azure Database Migration Service · Amazon Schema Conversion Tool · Google Cloud Database Migration Service · Azure Database Migration Service · Amazon Application Migration Service · Google Cloud Migrate for Compute Engine · Azure Migrate · Amazon Migration Hub · Google Cloud Migration Center · Azure Migrate · Amazon Application Discovery Service · Google Cloud Migration Center · Azure Migrate · Amazon Migration Evaluator · Google Cloud Migration Center · Azure Migrate · Amazon Cloud Adoption Framework · Google Cloud Adoption Framework · Azure Cloud Adoption Framework · Amazon Professional Services · Google Cloud Professional Services · Azure Professional Services · Amazon Partner Network · Google Cloud Partner Advantage · Azure Partner Network · Amazon Marketplace · Google Cloud Marketplace · Azure Marketplace · Amazon Support · Google Cloud Support · Azure Support · Amazon re:Post · Google Cloud Community · Azure Community · Amazon Training and Certification · Google Cloud Training · Azure Training · Amazon Skill Builder · Google Cloud Skills Boost · Azure Learn · Amazon Documentation · Google Cloud Documentation · Azure Documentation · Amazon Whitepapers · Google Cloud Whitepapers · Azure Whitepapers · Amazon Blogs · Google Cloud Blog · Azure Blog · Amazon Events · Google Cloud Events · Azure Events · Amazon Podcasts · Google Cloud Podcasts · Azure Podcasts · Amazon Videos · Google Cloud Videos · Azure Videos · Amazon Newsletters · Google Cloud Newsletters · Azure Newsletters · Amazon Forums · Google Cloud Forums · Azure Forums · Amazon Stack Overflow · Google Cloud Stack Overflow · Azure Stack Overflow · Amazon GitHub · Google Cloud GitHub · Azure GitHub · Amazon Open Source · Google Cloud Open Source · Azure Open Source · Amazon Contributions · Google Cloud Contributions · Azure Contributions · Amazon Roadmap · Google Cloud Roadmap · Azure Roadmap · Amazon Feedback · Google Cloud Feedback · Azure Feedback · Amazon Ideas · Google Cloud Ideas · Azure Ideas · Amazon Voting · Google Cloud Voting · Azure Voting · Amazon Status · Google Cloud Status · Azure Status · Amazon Service Health Dashboard · Google Cloud Service Health · Azure Service Health · Amazon Personal Health Dashboard · Google Cloud Personal Service Health · Azure Personal Health

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(7, 8, 9, 10, 11, 12)	8	12688	-0.119959	2.444875	-8.768727	-1.365874	-0.021789	1.066772	12.398527	48.9	51.1	alt_util_A_APPROVED_t10
(7, 8, 9, 10, 11, 12)	9	12688	-459.401827	651.956237	-3177.08958	-850.529082	-294.528789	-4.862259	1972.601016	24.4	75.6	res_112
(7, 8, 9, 10, 11, 12)	10	12688	-374.582741	540.435618	-3771.909569	-626.896187	-289.957738	-27.305925	1727.376903	22.3	77.7	hour
(7, 8, 9, 10, 11, 12)	11	12688	2.27997	412.396103	-1483.528237	-249.375494	-35.700394	202.490858	2162.018326	46.2	53.8	alt_util_A_CANCELLED_t12
(7, 8, 9, 10, 11, 12)	12	12688	0.413697	2.050001	-9.979079	-0.366165	0.234159	1.367636	14.098855	62.0	38.0	alt_util_A_REGISTERED_t9
(9, 10)	9	476	1.220177	0.55685	-0.114252	0.70271	1.382278	1.637644	2.161287	99.2	0.8	hour
(9, 10)	10	476	-0.949345	0.517322	-1.795174	-1.343675	-1.094743	-0.454462	0.273408	5.0	95.0	hour
(9, 11)	9	1055	-0.524243	0.212738	-1.156798	-0.663978	-0.54431	-0.426461	0.022907	0.5	99.5	hour
(9, 11)	11	1055	0.524168	0.212527	-0.022479	0.426704	0.544259	0.663661	1.156732	99.5	0.5	hour
(10, 11)	10	715	2.697526	0.488941	1.101782	2.384306	2.750475	3.038527	3.595585	100.0	0.0	hour
(10, 11)	11	715	-2.696851	0.489537	-3.599736	-3.039866	-2.748076	-2.387942	-1.100781	0.0	100.0	hour

Key Assumptions & Interpretation Notes

Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md